$$\tilde{\Theta}_3' = \frac{n+1}{n} x_{max}$$

Докажем состоятельность:

$$\forall \Theta > 0 \ \forall \varepsilon > 0 \ P(|\tilde{\Theta}_3' - \Theta| \geq \varepsilon) \longrightarrow 0$$

$$P\left(\left|\frac{n+1}{n} x_{max} - \Theta\right| \geq \varepsilon\right) =$$

$$= P\left(\frac{n+1}{n} x_{max} \geq \Theta + \varepsilon\right) + P\left(\frac{n+1}{n} x_{max} \leq \Theta - \varepsilon\right) =$$

$$= P\left(x_{max} \geq \underbrace{\frac{n}{n+1}(\Theta+\varepsilon)}_{\xrightarrow{n\to\infty} \Theta+\varepsilon}\right) + P\left(x_{max} \leq \frac{n}{n+1}(\Theta-\varepsilon)\right)$$

$$\Downarrow$$

$$P\left(x_{max} \geq \frac{n}{n+1}(\Theta+\varepsilon)\right) \xrightarrow[n\to\infty]{} 0 \quad (\text{т.к. } x_{max} \leq \Theta \text{ всегда})$$

$$P\left(x_{max} \leq \frac{n}{n+1}(\Theta-\varepsilon)\right) = \Psi\left(\frac{n}{n+1}(\Theta-\varepsilon)\right) = \left(\frac{\frac{n}{n+1}(\Theta-\varepsilon)}{\Theta}\right)^n =$$

$$= \left(\cancel{\frac{(\Theta-\varepsilon)}{n+1}}\right)^n = \left(\frac{n}{n+1}\right)^n \cdot \left(\frac{\Theta-\varepsilon}{\Theta}\right)^n = \left(\frac{1}{1+\frac{1}{n}}\right)^n \cdot \left(1-\frac{\varepsilon}{\Theta}\right)^n$$

$$\left.\begin{array}{l} \left(\frac{1}{1+\frac{1}{n}}\right)^n \xrightarrow[n\to\infty]{} \frac{1}{e} \\ \varepsilon > 0 \Rightarrow 0 < \frac{\varepsilon}{\Theta} < 1 \Rightarrow \left(1-\frac{\varepsilon}{\Theta}\right)^n \xrightarrow[n\to\infty]{} 0 \end{array}\right\} \Rightarrow P(|\tilde{\Theta}_3' - \Theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$$

(т.к. при $\varepsilon \geq \Theta$ сост. сразу)

$$\Downarrow$$

$\tilde{\Theta}_3'$ состоятельна по опр.

//

$$\tilde{\Theta}_3 = x_{max}$$

$$P(|x_{max} - \Theta| \geq \varepsilon) = \underbrace{P(x_{max} \geq \Theta + \varepsilon)}_{=0} + P(x_{max} \leq \Theta - \varepsilon)$$

(т.к. $x \in (0; \Theta)$)

$$P(x_{max} \leq \Theta - \varepsilon) = \Psi(\Theta - \varepsilon) = \left(\frac{\Theta - \varepsilon}{\Theta}\right)^n = \left(1 - \frac{\varepsilon}{\Theta}\right)^n \xrightarrow[n \to \infty]{} 0 \Rightarrow$$

оценка состоятельна по вер //